TELECOM CUSTOMER CHURN ANALYSIS

Executive Summary | Final corrected edition

7,043	1,869	26.54%	73.46%
Total customers	Churned customers	Overall churn	Retention

1. Project Overview

This project analyzes customer churn in a telecommunications dataset containing 7,043 customer records and 21 columns. The workflow combines data profiling and cleaning, SQL churn and revenue analysis, Python exploratory data analysis, statistical testing, and practical retention recommendations.

Business objective: identify customer groups with higher churn, quantify monthly charges associated with churned customers, and recommend focused retention actions supported by the data.

2. Data Quality and Preparation

The original TotalCharges field contained 11 whitespace-only values. All 11 belonged to customers with tenure = 0 and Churn = No. The blanks were converted to missing values and the column was converted to numeric. No rows were removed and no values were imputed.

Data check	Observed result
Rows / columns	7,043 / 21
Unique customer IDs	7,043
Duplicate customer IDs	0
Missing values after cleaning	11, all in TotalCharges
Other missing values	0
Tenure range	0–72 months

Data handling: the 11 missing TotalCharges values were retained because they correspond to zero-tenure customers; filling them with an assumed amount would introduce unsupported data.

3. Overall Churn and Main Segment Findings

Of the 7,043 customers, 1,869 churned (26.54%) and 5,174 were retained (73.46%). Segment churn rates below are calculated within each segment.

Segment	Customers	Churned	Churn rate
Month-to-month contract	3,875	1,655	42.71%
One-year contract	1,473	166	11.27%
Two-year contract	1,695	48	2.83%
New tenure (0–12 months)	2,186	1,037	47.44%
Early tenure (13–24 months)	1,024	294	28.71%

Segment	Customers	Churned	Churn rate
Mid-term tenure (25–48 months)	1,594	325	20.39%
Long-term tenure (49–72 months)	2,239	213	9.51%
Electronic check	2,365	1,071	45.29%
Fiber optic internet	3,096	1,297	41.89%

Key finding: new month-to-month customers are a high-priority overlap: 1,994 customers, 1,024 churned, and a 51.35% churn rate. This group represents 54.79% of all churned customers.

4. Additional Customer Patterns

Characteristic	Group	Churn rate
Internet service	Fiber optic	41.89%
Internet service	DSL	18.96%
Internet service	No internet service	7.40%
Payment method	Electronic check	45.29%
Payment method	Mailed check	19.11%
Payment method	Bank transfer (automatic)	16.71%
Payment method	Credit card (automatic)	15.24%
Paperless billing	Yes	33.57%
Paperless billing	No	16.33%
Senior citizen	Yes	41.68%
Senior citizen	No	23.61%

Customers without OnlineSecurity had a 41.77% churn rate, compared with 14.61% among customers with the service. For TechSupport, the corresponding rates were 41.64% without the service and 15.17% with it. These are associations, not proof that adding a service will prevent churn.

5. Revenue Exposure Among Churned Customers

The sum of MonthlyCharges for the 1,869 churned customers is $139,130.85, averaging $74.44 per churned customer. Multiplying by 12 gives an illustrative annualized run-rate of $1,669,570.20.

Segment	Churned customers	Monthly charge exposure	Share of exposure
Month-to-month	1,655	$120,847.10	86.86%
One year	166	$14,118.45	10.15%
Two year	48	$4,165.30	2.99%
New tenure	1,037	$68,954.25	49.56%
Fiber optic	1,297	$114,300.05	82.16%
Electronic check	1,071	$84,288.75	60.58%

Financial limitation: these figures describe monthly charges attached to customers recorded as churned. They are not verified realized revenue losses or a forecast. The annualized value is a simple 12× illustration. TotalCharges is historical accumulated billing, not monthly recurring revenue.

6. Exploratory Analysis — Numerical Variables

Measure	Retained customers	Churned customers
MonthlyCharges — mean	$61.27	$74.44
MonthlyCharges — median	$64.43	$79.65
Tenure — mean	37.57 months	17.98 months
Tenure — median	38 months	10 months
TotalCharges — mean	$2,555.34	$1,531.80
TotalCharges — median	$1,683.60	$703.55

Churned customers tend to have shorter tenure and higher monthly charges. Their accumulated TotalCharges are lower, consistent with shorter observed tenure. TotalCharges summaries exclude the 11 missing values, all from retained customers.

7. Statistical Testing

The statistical notebook runs eight Chi-square tests for categorical variables and three Mann–Whitney U tests for numerical variables. Holm correction is applied across all 11 tests. All eight Chi-square tables passed the expected-frequency check: every expected count was at least 5. All 11 tests remained statistically significant after Holm correction at $\alpha = 0.05$.

Categorical variable	Raw p-value	Strength of association (Cramér's V)
Contract	5.863038 × 10■²■■	0.4101
Tenure group	2.943770 × 10■¹■■	0.3486
OnlineSecurity	2.661150 × 10■¹■■	0.3474
TechSupport	1.443084 × 10■¹■■	0.3429
InternetService	9.571788 × 10■¹■■	0.3225
PaymentMethod	3.682355 × 10■¹■■	0.3034
PaperlessBilling	4.073355 × 10■■■	0.1915
SeniorCitizen	1.510067 × 10■³■	0.1505

Mann–Whitney U comparisons

Numerical variable	Retained median	Churned median	Raw p-value
Tenure	38	10	2.419636 × 10■²■■
MonthlyCharges	$64.43	$79.65	3.311628 × 10■■■
TotalCharges	$1,683.60	$703.55	1.995985 × 10■■■

Contract had the strongest categorical association with churn (Cramér's V = 0.4101). Statistical significance indicates evidence of association or distributional differences in this sample; it does not establish causation.

8. Retention Recommendations

Priority area	Recommended action	Suggested KPIs / evaluation
New customers, especially month-to-month	First-90-day check-ins, early complaint detection, clear support communication, and targeted retention offers	90-day retention; new-customer churn; complaint resolution time; longer-co
Fiber optic customers	Review speed and reliability complaints, monitor service issues and improve technical support response	Fiber churn; unresolved complaints; resolution time; satisfaction. Test with a
Electronic-check customers	Investigate payment failures and billing confusion; improve reminders and explain payment alternatives	Electronic-check churn; payment failures; billing complaints; successful pay

These are hypotheses to test, not proven causes. The dataset does not contain direct measures of speed, outages, payment failures, complaints or satisfaction. A controlled pilot is needed to measure whether these actions improve retention.

9. Project Workflow and Deliverables

Workstream	Deliverable / method
Data understanding and cleaning	01_data_understanding notebook; cleaned CSV; type conversion and missing-value review
SQL profiling and quality checks	01_data_profiling.sql and 02_data_cleaning.sql
SQL churn analysis	03_churn_analysis.sql — churn by contract, tenure, payment, internet and services
SQL revenue analysis	04_revenue_analysis.sql — monthly charge exposure by segment
Python EDA	02_eda.ipynb — descriptive analysis and 15 exported charts
Statistical testing	03_statistical_testing notebook — Chi-square, Mann–Whitney U and Holm correction
Retention strategy	04_retention_strategy.ipynb — three initiatives, KPIs and pilot plans

10. Limitations and Conclusion

This is an observational analysis of one customer dataset. It identifies segment patterns and associations but cannot establish why customers left. It does not include a time-based churn forecast, causal experiment, direct service-quality measures, complaint records or confirmed realized revenue loss. Segment dimensions overlap, so customer counts and exposure values must not be added across different segment analyses.

The clearest priorities are early-tenure and month-to-month customers, fiber optic customers, and electronic-check customers. The next business step is to test targeted retention actions with control groups and measure actual retention and financial outcomes before scaling them.

Prepared using the supplied SQL scripts, notebooks and executed outputs, cleaned dataset, and 15 EDA visualizations.